

An Analytic Two-Dimensional Counterexample to the Wronski-Basis Conjecture

Candidate counterexample for arXiv:2508.04639
Run `fa_banach_001`, lane 7

12 August 2026

Status. Candidate full counterexample, likely valid, pending specialist human review. It disproves Conjecture 8 already for a two-dimensional Hilbert space of entire functions.

1 Source question

Micheas asks whether every orthonormal basis of a Hilbert function space is the fundamental solution set, via Wronski orthogonalization, of a unique linear homogeneous ordinary differential equation (HODE), possibly of infinite order [1, Conjecture 8, p. 16].

Based on the theoretical development in this section, we can see that the Wronski orthogonalization process when applied to ODEs, provides a set of linearly independent and orthogonal set of functions $\mathcal{F}_N$ as the fundamental set of solutions of a HODE. Combining these two important properties makes $\mathcal{F}_N$ a basis for the spanned space $\overline{\mathcal{F}}_N$ of $\mathcal{F}_N$. Therefore, this observation leads to the following general conjecture regarding bases functions in Hilbert function spaces.

Conjecture 8 (Bases as solutions to HODEs) *Let $(\mathcal{F}_M^{\mathbb{M}}, \rho)$ be a Hilbert function space. Assume that $\mathcal{E} = \{e_i\}_{i \in I} \subset \mathcal{F}_M^{\mathbb{M}}$, with $\overline{\mathcal{E}} = \mathcal{F}_M^{\mathbb{M}}$, is an orthonormal basis of $\mathcal{F}_M^{\mathbb{M}}$, where the index set I is countable or uncountable. Then, there exists a unique linear HODE (of potentially infinite order) such that its fundamental set of solutions via Wronski orthogonalization is the set $\mathcal{E}$.*

Since the basis function set $\mathcal{E}$ is constructed via Wronski orthogonalization, it will be aptly called the Wronski basis of $\mathcal{F}_M^{\mathbb{M}}$. In view of Theorem 5, Remark 6 and Example 7,

Figure 1: Conjecture 8 in arXiv:2508.04639v4, PDF page 16.

Immediately after the conjecture, the paper says that it is satisfied “trivially” for a finite basis $\{e_1, \dots, e_N\}$ and identifies the claimed equation as the N th-order Wronskian quotient

$$\frac{W(e_1, \dots, e_N, y)}{W(e_1, \dots, e_N)} = 0.$$

The paper’s regular-ODE convention requires continuous coefficients on the domain and a leading coefficient that is nonzero at every point [1, Remark 1]. We use exactly that finite-dimensional interpretation.

2 Proof intuition

Linear independence of functions does not imply that their Wronskian is pointwise nonzero. This distinction is invisible if one divides formally by the Wronskian. The smallest possible example is $1, x^2$: they are linearly independent and can be made an orthonormal basis, but

$W(1, x^2) = 2x$ vanishes at the origin. The formal Wronskian equation is $xy'' - y' = 0$, which is necessarily singular at that point. Direct substitution shows that no other regular second-order equation can avoid the same obstruction.

3 Counterexample

Let

$$\mathcal{H} = \text{span}_{\mathbb{R}}\{1, x^2\}$$

as a space of real-valued functions on $\mathbb{R}$, and define

$$\langle a + bx^2, c + dx^2 \rangle_{\mathcal{H}} = ac + bd.$$

Theorem 1. *The space $\mathcal{H}$ is a Hilbert function space and $\mathcal{E} = \{1, x^2\}$ is an orthonormal basis of entire functions. There is no regular linear second-order HODE on $\mathbb{R}$ having $\mathcal{E}$ as a fundamental set of solutions. Consequently, Conjecture 8 and its asserted finite-basis case are false as stated.*

Proof. The coefficient map

$$\mathbb{R}^2 \longrightarrow \mathcal{H}, \quad (a, b) \longmapsto a + bx^2,$$

is an isometric linear bijection. Thus $\mathcal{H}$ is complete, and $\{1, x^2\}$ is an orthonormal basis. Both functions are entire.

Suppose that a regular second-order linear HODE on $\mathbb{R}$,

$$p_2(x)y''(x) + p_1(x)y'(x) + p_0(x)y(x) = 0, \tag{1}$$

has 1 and x^2 as solutions. Here the coefficients are continuous and $p_2(x) \neq 0$ for every $x \in \mathbb{R}$, as required by the source paper's regularity convention. Substituting $y = 1$ into (1) gives

$$p_0(x) = 0 \quad (x \in \mathbb{R}).$$

Substituting $y = x^2$ then gives

$$2p_2(x) + 2xp_1(x) = 0 \quad (x \in \mathbb{R}).$$

At $x = 0$ this says $p_2(0) = 0$, contradicting regularity. Hence no such HODE exists.

A fundamental set for a regular N th-order equation has N elements; this is also the convention used in the source paper's finite-basis paragraph. Thus the two-element basis is precisely the $N = 2$ case claimed there, and failure of existence already disproves the universal conjecture. $\square$

4 The failed Wronskian formula

The source paper's proposed formula makes the singularity completely explicit:

$$W(1, x^2) = \det \begin{pmatrix} 1 & x^2 \\ 0 & 2x \end{pmatrix} = 2x,$$

whereas expansion of the augmented determinant gives

$$W(1, x^2, y) = \det \begin{pmatrix} 1 & x^2 & y \\ 0 & 2x & y' \\ 0 & 2 & y'' \end{pmatrix} = 2xy'' - 2y'.$$

Away from the origin, division by $W(1, x^2)$ produces $y'' - x^{-1}y' = 0$. On all of $\mathbb{R}$, its undivided form $xy'' - y' = 0$ has leading coefficient x , so it is singular exactly where the denominator

vanishes. Reordering or rescaling the basis only multiplies the Wronskian by a nonzero constant and cannot repair this zero.

More generally, if two C^2 functions form a fundamental system for a regular second-order equation on an interval, Abel’s identity forces their Wronskian either to vanish nowhere or to vanish identically. Therefore any linearly independent pair with a Wronskian having an isolated zero supplies the same kind of obstruction.

5 Verification notes

There are three independent exact checks.

1. The coefficient map above proves completeness and orthonormality.
2. Direct substitution into an arbitrary regular equation forces $p_2(0) = 0$.
3. Direct determinant expansion yields $W(1, x^2) = 2x$ and $W(1, x^2, y) = 2xy'' - 2y'$.

No numerical computation or unproved external lemma is needed.

6 Scope, repair, and novelty check

The counterexample concerns the conjecture exactly as stated on the global domain $\mathbb{R}$ and its own finite-order interpretation. It does not rule out a corrected theorem on an interval avoiding the origin: on $(0, \infty)$, for example, $y'' - x^{-1}y' = 0$ is regular. Nor does it address a newly defined notion of “fundamental set” for infinite-order equations. A necessary finite-dimensional repair is to require the basis Wronskian to be nonzero everywhere on the domain; under the usual differentiability assumptions, the standard Wronskian formula then applies.

On 12 August 2026, bounded arXiv searches for the exact title, the exact phrase “Bases as solutions to HODEs”, and “Wronski basis” found the source paper but no later arXiv paper or note answering Conjecture 8. This is a bounded novelty check, not a claim of exhaustive bibliographic priority.

7 Human review recommendation

Review as a likely valid full counterexample. The two points worth checking are that finite-dimensional function spaces are included in the source’s phrase “any Hilbert function space” (as its finite-basis paragraph itself indicates), and that its stated regularity convention is imposed globally on $\mathbb{R}$. Once those are accepted, the contradiction is elementary.

References

- [1] Athanasios Christou Micheas, *Construction of linearly independent and orthogonal functions in Hilbert function spaces via Wronski determinants*, arXiv:2508.04639v4, 2026. <https://arxiv.org/abs/2508.04639>.